

# PHASE 4 – DEVELOPMENT & IMPLEMENTATION

Project: LegalEase – AI-Powered Legal Document Generator

## 1. Core Functionality

The main functionality is legal document generation based on document type, parties, terms, and dates.

## 2. Backend Implementation

FastAPI is initialized in main.py. A root endpoint verifies that the service is running, and routes.py contains the document-generation logic. A Pydantic DocumentRequest model validates incoming data.

## 3. AI Implementation

GeminiDocumentGenerator is initialized and used to generate legal content from structured user input.

## 4. Frontend Implementation

Streamlit is used to create the user interface. Input fields collect document type, parties, terms, and dates. Generated content is displayed for preview and editing.

## 5. Export Implementation

The system supports TXT, DOCX, and PDF downloads. DOCX formatting includes logo, headings, terms table, and footer. PDF formatting includes logo, headings, bullets, header/footer, and professional layout.

## 6. Utility Functions

The source documentation identifies `sanitize_text(text)`, `format_docx(text, doc_type)`, `format_pdf(text, doc_type)`, and `format_html_preview(text)` as core utilities.

Source basis: LegalEase project documentation supplied for this task.